



SUPERIOR UNIVERSITY

Information Engineering Technology

Project: Room Availability

Subject: Software Engineering

SUBMITTED TO:

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Software Requirements Specification (SRS)

1. Introduction:

The University Campus Room Availability Management System is a software application developed to manage and monitor room availability inside a university campus. Universities contain multiple classrooms, laboratories, seminar halls, meeting rooms, conference rooms, and activity spaces that are used daily by students, faculty members, and administrative staff. Managing these rooms manually often creates problems such as double booking, scheduling conflicts, incorrect room allocation, lack of communication, and inefficient resource utilization.

The purpose of this project is to create a centralized computerized system that allows users to search available rooms, reserve rooms, cancel reservations, and manage schedules in real time. The system will improve operational efficiency and help the university administration utilize campus resources effectively.

This system aims to:

- Reduce manual paperwork
- Eliminate scheduling conflicts
- Provide accurate room availability information
- Improve communication between departments
- Save time for students and faculty
- Increase transparency in room allocation

2. Purpose of the Project:

The purpose of the **University Campus Room Availability Management System** is to develop an efficient, reliable, and automated software solution for managing room availability and reservations within a university campus. Universities contain a large number of classrooms, laboratories, seminar halls, conference rooms, meeting rooms, and activity spaces that are used daily by students, teachers, administrative staff, and different departments. Managing these rooms manually often creates many operational problems such as scheduling conflicts, double booking, incorrect room allocation, lack of communication, and wastage of university resources. Therefore, this system is designed to replace traditional manual processes with a modern computerized management system.

The primary purpose of this project is to provide a centralized platform where users can easily check room availability, reserve rooms, cancel reservations, and view schedules in real time. The system helps university administration manage campus facilities more effectively while reducing paperwork and human errors. By introducing automation, the university can improve overall operational efficiency and ensure better utilization of available rooms and facilities.

Currently, in many universities, room scheduling is handled using paper records, spreadsheets, notice boards, or manual communication between departments. These methods are time-consuming and unreliable because multiple departments may request the same room at the same time, leading to confusion and conflicts. Sometimes classrooms remain unused because there is no proper monitoring system to identify available spaces. Faculty members and students also face difficulties when searching for vacant rooms for lectures, meetings, seminars, presentations, workshops, or study sessions. The proposed system aims to solve these problems by maintaining accurate and updated records of room availability.

3. Scope of the Project:

The scope of the **University Campus Room Availability Management System** defines the boundaries, functionalities, features, and operations that will be included in the project. This system is designed to manage and monitor room availability, scheduling, and reservations within the university campus environment. The software will provide a centralized platform that allows students, faculty members, administrative staff, and university management to access room-related information and perform booking operations efficiently.

- Classrooms
- Computer laboratories
- Science laboratories
- Seminar halls
- Conference rooms
- Meeting rooms
- Auditorium halls
- Training rooms
- Faculty discussion rooms

4.Elicitation:

Elicitation is one of the most important phases in software development and system analysis. It is the process of gathering, understanding, analyzing, and identifying the requirements of users, stakeholders, and organizations for a software system. In the University Campus Room Availability Management System, elicitation helps developers understand what features, functions, and services are needed by students, faculty members, university administrators, and IT staff.

- Problems in the current manual room management process
- User expectations and requirements
- Functional and non-functional requirements
- Security and performance needs

5.Interviews:

Structured interviews included pre-planned questions prepared before the interview session. These interviews focused on collecting specific information related to room booking, scheduling, and system requirements

Interview Questions for Students:

General Questions

- How do you currently find available classrooms or rooms on campus?
- Do you face any difficulties in locating empty rooms for study or group work?
- How often do you need to use campus rooms outside your scheduled classes?
- Are you satisfied with the current room booking or allocation system? Why or why not?
- What problems do you usually face when trying to access a room?

Room Availability Questions

- Do you think it is easy to know which rooms are free at a specific time?
- Have you ever experienced a situation where you went to a room and found it already occupied?
- How important is real-time room availability information for you?
- Would you prefer a system that shows all available rooms instantly? Why?
- What information should be shown when checking room availability (e.g., capacity, equipment, location)?

Booking System Questions

- Do you think students should be allowed to book rooms online?
- Have you ever tried booking a room through the administration office? How was your experience?
- What problems do you face in the current manual booking process?
- How simple should the online booking process be for students?
- Should students be able to cancel or modify their bookings? Why?

System Features Questions

- Which features would you like in a room management system?
- Do you think filters like building name, room size, and equipment are useful?
- Would you prefer a search system or a calendar-based system for booking rooms?
- Should the system send notifications about booking confirmation or changes?
- What additional features would make the system more useful for students?

Usability and Accessibility Questions

- How important is it for you that the system is easy to use?
- Would you prefer accessing the system through mobile phones or computers?
- Do you think the system should be available 24/7? Why?
- What difficulties do you face when using university-related software systems?
- How can the interface be improved to make it more student-friendly?

Performance and Expectations Questions

- How fast do you expect room availability results to appear?
- Would delays in system response affect your usage?
- How reliable do you expect the system to be during exam periods or busy hours?
- What problems occur when many students try to use rooms at the same time?
- What improvements would make your academic life easier regarding room usage?

Open-Ended Questions

- What features do you think are missing in the current room management system?
- If you could design this system yourself, what would it look like?
- What are the biggest challenges you face regarding room availability?
- How do you think this system can improve university life?
- Any suggestions or ideas to improve room booking and availability management?

4. Requirement Analysis

Requirement analysis was performed to study the collected requirements and identify the most important features needed for the system.

Prioritization:

Prioritization was performed to identify and arrange the most important features required in the **University Campus Room Availability Management System**. The purpose of prioritization is to ensure that the system is developed in a structured way, where essential functions are implemented first, followed by additional enhancements based on importance and feasibility.

- Student and faculty needs
- Frequency of system usage
- Importance in solving real problems (e.g., room conflicts)
- Improvement of communication and coordination
- Overall project scope and limitations
- Technical feasibility and development complexity

High Priority Features

- User authentication and login system
- Room availability search and filtering
- Real-time room scheduling and updates
- Online room booking system
- Conflict detection (prevents double booking)
- Admin management of rooms and schedules
- Database system for storing all records
- User role management (Student, Faculty, Admin)

Medium Priority Features

- Booking cancellation system
- Notification system (email/system alerts for bookings)
- Booking history and record tracking
- Admin reporting and monitoring dashboard
- Advanced search filters (equipment, department, etc.)
- User profile management system

Low Priority Features

- Mobile application support
- AI-based room recommendations
- QR code-based room access system
- Voice assistant support
- Multi-language support

Conflict Resolution:

Conflict resolution is an important part of the **University Campus Room Availability Management System**. It ensures that no two users can reserve the same room at the same time and helps maintain an organized and error-free scheduling process. In university environments, conflicts usually occur when multiple departments, faculty members, or student groups try to book the same room for overlapping time slots. This system is designed to automatically detect and resolve such conflicts.

5. Functional Requirements

Functional requirements describe the functions and services the system should provide.

Room Availability System should be able to:

- The system shall allow users to log in using valid university credentials.
- The system shall verify username and password before granting access.
- The system shall assign roles such as Student, Faculty, and Admin.
- The system shall deny access to unauthorized users.

Functional Requirement Results

The functional requirement results describe the **outcome and expected behavior** of the system after implementing all functional requirements. These results show how the system will operate in real-world usage and what users will experience when interacting with it.

User Authentication Result:

- Users are able to successfully log in using university credentials.
- The system correctly identifies user roles (Student, Faculty, Admin).
- Unauthorized users are blocked from accessing the system.
- Secure login and logout sessions are maintained without data leakage.

User Management Result

- Admin can successfully create, update, and delete user accounts.
- User information is stored correctly in the database.
- Role-based access is applied properly for each user type.
- Password reset functionality works correctly when required.

Room Search Result

- Users can successfully search rooms using filters such as date, time, and capacity.
- The system displays accurate and relevant search results.
- Search results update instantly based on user input.

Room Booking Result

- Users can book available rooms successfully.
- The system prevents double booking and scheduling conflicts.
- Each booking generates a unique booking ID.
- All booking data is stored securely in the database.

6. Non-Functional Requirements:

Non-functional requirements define the **quality attributes and system constraints** of the University Campus Room Availability Management System. Unlike functional requirements, they do not describe specific system functions but instead focus on how well the system performs, how secure it is, how reliable it is, and how easy it is to use and maintain.

- Provide fast response time
- Have a user-friendly interface
- Ensure secure handling of student information
- Provide reliable and accurate responses
- Support continuous system availability
- Be easy to maintain and update
- Work efficiently on different devices
- Provide stable system performance
- Reduce system downtime
- Ensure data privacy and security

Non-Functional Requirement Results

The non-functional requirements improve system quality, security, usability, and overall user experience.

6. System Requirements

Software Requirements

- HTML5 for page structure
- CSS3 for styling
- JavaScript for dynamic functionality

Hardware Requirements

- Laptop or Desktop Computer

7. Validation

Requirement validation is the process of checking whether the collected requirements are correct, realistic, complete, and achievable for the project.

The main purposes of validation are:

- To ensure the system meets user requirements
- To confirm that all features work correctly
- To verify that the system is suitable for real university use
- To identify missing or incorrect requirements
- To improve system quality before final deployment

1. Requirements Traceability Matrix (RTM)

	A	B	C	D	E	F	G
1	Req_ID	Requirement Description	Source	Design Module	Implementation	Test Case	Status
2	R1	User login and authentication system	Students/Admin	Login Module	Authentication API	TC01	Completed
3	R2	Role-based access control	Admin	Security Module	RBAC System	TC02	Completed
4	R3	Room search functionality	Students	Search Module	Search Engine	TC03	Completed
5	R4	Real-time room availability display	Students/Faculty	Availability Module	Scheduling System	TC04	Completed
6	R5	Room booking system	Faculty/Admin	Booking Module	Booking API	TC05	Completed
7	R6	Booking cancellation feature	Students/Faculty	Booking Module	Cancellation API	TC06	Completed
8	R7	Conflict detection system	Admin	Scheduling Module	Conflict Handler	TC07	Completed
9	R8	Admin room management	Admin	Admin Panel	CRUD Operations	TC08	Completed
10	R9	Notification system	Students/Faculty	Notification Module	Email/SMS API	TC09	Completed
11	R10	Reporting system	Admin	Reports Module	Analytics Engine	TC10	Completed
12	R11	Database management system	System	Database Layer	MySQL/PostgreSQL	TC11	Completed
13	R12	Dashboard interface	All Users	UI Module	Frontend UI	TC12	Completed
14							

11. System Features

Main Functions

- Answer student questions automatically
- Provide quick responses
- Support 24/7 communication
- Reduce manual workload on staff

Room Availability Feature

The system allows students, faculty, and staff to check room availability in real time through the system. This feature helps users easily find available classrooms, labs, and seminar halls without visiting the administration office. It reduces scheduling conflicts and improves room utilization across the campus.

Main Functions:

- Display real-time room availability status
- Show booked and available time slots for each room
- Allow users to check rooms by selecting date and time
- Provide detailed room information such as capacity, location, and facilities
- Update availability automatically after booking or cancellation

3. Room Booking Feature

This feature allows authorized users to reserve available rooms for academic and administrative purposes. It ensures that rooms are allocated properly without conflicts or double booking.

Main Functions:

- Online room reservation system
- Select date, time, and duration for booking
- Automatic conflict detection
- Generate booking confirmation with unique ID
- Store booking details in the database

4. Room Search System

The system provides a search feature to help users quickly find suitable rooms based on their requirements.

Main Functions:

- Search rooms by date and time
- Filter rooms by building, capacity, and equipment
- Display list of available rooms
- Provide instant search results

5. Notification System

The system sends notifications to users regarding booking status and updates. This improves communication and ensures users stay informed.

Main Functions:

- Send booking confirmation alerts
- Notify about booking cancellation
- Provide schedule change updates
- Display system notifications in dashboard

6. Admin Management Feature

The admin feature allows university staff to manage rooms, users, and bookings in the system.

Main Functions:

- Add, update, and delete room information
- Manage user accounts and roles
- Approve or reject booking requests
- Monitor system activities

7. User-Friendly Interface

The system provides a simple and easy interface for all users to interact with room availability and booking functions.

Main Functions:

- Easy navigation system
- Simple search and booking process
- Responsive design for all devices
- Quick access to room information

8. Secure Information Management

The system ensures that all room and user data is stored securely and accessed only by authorized users.

Main Functions:

- Secure login system
- Role-based access control
- Data protection and privacy
- Safe storage of booking records

9. Conclusion

The **University Campus Room Availability Management System** is designed to improve room management and scheduling within the university. It provides real-time room availability, online booking, conflict detection, and secure data management.

This system reduces manual work, prevents scheduling conflicts, and improves the efficiency of campus resource utilization. The requirement engineering process helped identify all essential features needed to build an effective and user-friendly room management system.

The final SRS document clearly defines the system scope, requirements, and functionalities, providing a strong foundation for development and implementation.

